

Figure F2: L1 Wide-Format Column Structure to Normalised Long-Form Schema Ingestion

Wide-Format Table: Raw Chandrayaan-3 L1 CSV (2,055 Columns)							
Housekeeping & Metadata (6 Columns)				Spectral Wavelength Intensity Channels (2,049 Columns)			
Time	Count	Reset	Fire	164.35 nm	164.74 nm	...	878.26 nm
11:35:22	1	1	0	366.0	384.0	...	305.0

Melt / Wide-to-Long Reshape Ingestion Invariant



Normalized Long-Format Table: spectral_data Schema (Rows × 2,094 per shot)					
Measurement_ID (PK)	Wavelength_nm (PK)	Cleaned_Intensity	Laser_Energy_V	Time_UTC	
1	164.35	0.0	3327	11:35:22	
1	164.74	0.0	3327	11:35:22	
...	...	...	...	...	
1	878.26	3.0	3327	11:35:22	